

The combined effects of healthy lifestyle behaviors on all cause mortality: a systematic review and meta-analysis.

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OBJECTIVE: Lifestyle factors are related to mortality. Although much is known about the impact of single factors, the current evidence about the combined effects of lifestyle behaviors on mortality has not yet been systematically compiled. METHOD: We searched Medline, Embase, Global Health, and Samed up to February 2012. Prospective studies were selected if they reported the combined effects of at least three of five lifestyle factors (obesity, alcohol consumption, smoking, diet, and physical activity). The mean effect sizes that certain numbers of combined lifestyle factors have on mortality were compared to the group with the least number of healthy lifestyle factors by meta-analysis. Sensitivity analyses were conducted to explore the robustness of the results. RESULTS: 21 studies (18 cohorts) met the inclusion criteria of which 15 were included in the meta-analysis that comprised 531,804 people with a mean follow-up of 13.24 years. The relative risks decreased proportionate to a higher number of healthy lifestyle factors for all cause mortality. A combination of at least four healthy lifestyle factors is associated with a reduction of the all cause mortality risk by 66% (95% confidence interval 58%-73%). CONCLUSION: Adherence to a healthy lifestyle is associated with a lower risk of mortality. Copyright © 2012. Published by Elsevier Inc.